

AIGR Gene Hypothesis Deep Research —

C. elegans STA-2 (UniProt Q20977)

Focus: function_assignment · Hypothesis slug `sta2-dbd-dna-binding` **Term under test:** GO:0000978 — *RNA polymerase II cis-regulatory region sequence-specific DNA binding* **Source:** `genes/worm/sta-2/sta-2-ai-review.yaml` (free-text)

Executive Judgment

Verdict: Partially supported — the STAT fold is retained, but functional sequence-specific DNA binding (GO:0000978) is NOT supported (likely homology-driven over-annotation).

The seed hypothesis has two separable claims. The first — that STA-2 *retains a STAT DNA-binding-domain fold* — is **supported**. The second — that this fold is *capable of sequence-specific DNA binding* and therefore justifies GO:0000978 — is **unresolved-to-refuted** on the available evidence and should be treated conservatively by curators.

Structurally, the AlphaFold model of STA-2 (AF-Q20977-F1) reproduces the STAT DBD fold with high confidence (mean pLDDT $\approx$ 88 over the DBD region), and its DBD-region backbone deviates from the DNA-bound human STAT1 crystal (PDB 1BF5) by essentially the same amount (8.2 Å) as the ChIP-validated nematode binder STA-1 (8.0 Å). So the *scaffold* is present. However, every sequence-level and functional test points away from autonomous sequence-specific DNA binding: STA-2 is the **most divergent** STAT examined (only ~20% DBD identity to STAT1), it **uniquely fails** the Pfam STAT-DBD profile HMM (PF02864) that human STAT1, Drosophila Stat92E and C. elegans STA-1 all pass, it **conserves none** of the three base-contacting (sequence-specificity-determining) residues of STAT1 (E421, S459, N460), and it **lacks the canonical activating tyrosine** (STAT1 Y701) required for the phospho-dimerization that drives high-affinity STAT–DNA binding. No direct DNA-binding assay (EMSA, ChIP-seq, or defined consensus motif) exists for STA-2 in the primary literature; its function as a transcriptional activator of epidermal antimicrobial peptide (AMP) genes is established genetically and by localization, and it physically partners with the SLC6 transporter SNF-12 in an explicitly "unorthodox" mode of STAT regulation.

Most important caveats: (1) Fold conservation does not establish DNA-binding chemistry — it is necessary but not sufficient. (2) The absence of a direct binding assay is an *absence of evidence*, not proof of no binding; STA-2 could bind DNA in a partner-assisted or low-specificity manner. (3) The rigid-body RMSD values are crude (no outlier trimming) and should be read comparatively, not absolutely. The appropriate curation posture is to treat GO:0000978 on STA-2 as a homology-propagated prediction that is **less precise than the direct experimental knowledge** (nuclear transcriptional-activator role + SNF-12 interaction) and to generalize or remove it pending a direct assay.

Key Findings

F001 — STA-2 lacks the Pfam STAT DNA-binding-domain HMM that its DNA-binding paralog STA-1 retains

An InterPro/Pfam domain scan cleanly separates STA-2 from the canonical STATs and from its own nematode paralog. Human STAT1 (P42224), *Drosophila* Stat92E (Q24151) and *C. elegans* **STA-1 (Q9NAD6)** all match **Pfam PF02864 "STAT protein, DNA binding domain"** (STA-1 at residues 204–356; also the CDD cd14801 STAT-DBD model). ***C. elegans* STA-2 (Q20977) does NOT match PF02864** or the STAT-DBD CDD model. STA-2 hits only the weaker *fold-level* superfamilies — IPR008967 (p53-like transcription-factor DNA-binding superfamily, 164–443) and IPR012345 (STAT DNA-binding N-terminal superfamily, 167–313) — while retaining a clear SH2 domain (cd09919/PS50001, 449–526) and a divergent STAT-b N-terminal domain (PF24629, 27–163). STA-2 is additionally N-terminally truncated (567 aa versus 706–770 aa for canonical STATs) and lacks the coiled-coil (PF01017) and N-terminal interaction (PF02865) domains present in canonical STATs.

The interpretation is pivotal: a profile HMM like PF02864 is tuned to the conserved residues that *define* a functional STAT DBD. The fact that the ChIP-validated binder STA-1 passes it while STA-2 fails it — within the same organism and the same gene family — is a strong, discriminating signal that STA-2's DBD has diverged beyond the family's functional consensus, even though a fold-level signal persists.

F002 — STA-2 conserves the STAT DBD fold (AlphaFold pLDDT $\approx$ 88) but not the DNA-contacting residues of STAT1

Structural contact mapping on the DNA-bound STAT1 crystal (PDB **1BF5**) identified **16 STAT1 residues within 3.6 Å of DNA**, of which **3 make base-specific (sequence-reading) contacts: E421, S459, N460**. Pairwise Needleman–Wunsch (BLOSUM62) alignment of STAT1 to each ortholog, mapped onto these positions, gives a clear gradient of conservation:

Protein	Base-contacting (of 3) identical	BLOSUM- similar	All DNA- contacting (of 16) identical	DBD-region identity (STAT1 320–490)
Human STAT3	2/3	3/3	11/16	66%
Drosophila Stat92E	1/3 (E421 conserved)	—	5/16	30%
C. elegans STA-1 (ChIP+ control)	0/3	1/3	4/16	26%
C. elegans STA-2	0/3	0/3 (E421→S, S459→G, N460→P)	3/16	20% (most divergent)

STA-2 is the most divergent protein on every metric and uniquely loses *all* base-contacting residues with no conservative substitution (E421→S, S459→G, N460→P — charge/shape-disruptive changes at exactly the positions that read DNA sequence). Notably, even the positive-control binder STA-1 conserves 0/3 base-contacting residues identically — a reminder that nematode STATs read DNA through a diverged interface and that residue-identity at human positions is an imperfect proxy. The distinguishing feature for STA-2 is therefore the **combination** of (a) failing PF02864, (b) losing all base contacts without conservative substitution, and (c) being the overall most divergent DBD.

Despite this sequence divergence, AlphaFold (AF-Q20977-F1, v6) predicts the STA-2 DBD-region fold with high confidence: **mean pLDDT 88.1 over residues 164–443** (90% of residues > 70), comparable to the STA-1 DBD (81.4) and to STA-2's own SH2 domain (95.0). The scaffold is intact; the DNA-reading chemistry is not conserved.

F003 — STA-2 is a transcriptional activator of epidermal antimicrobial-peptide genes, but direct sequence-specific DNA binding is not demonstrated

Primary literature firmly establishes STA-2 as a STAT-like transcription factor controlling *C. elegans* epidermal antimicrobial peptide (AMP; e.g. *nlp-29*) gene expression — but always through genetics, localization, and target-gene readouts, never through a direct DNA-binding assay. Zhang et al. 2015 (PMID: 25692704) show STA-2 is tethered to apical hemidesmosomes and that architectural damage causes "*detachment of STA-2 molecules from hemidesmosomes and transcription of AMPs*," with STA-2 release inducing an innate immune response. Zhang et al. 2021 (PMID: 34166401) describe fungal effectors that act by "*preventing the key STAT-like transcription factor STA-2 from activating defensive antimicrobial peptide gene expression*," and report that one effector increases STA-2 levels in the nucleus.

Crucially, none of these studies report an EMSA, ChIP-seq, or a defined sequence motif for STA-2. The transcriptional-activator role is *inferred* from genetic epistasis, nuclear localization, and downstream target expression. The seed hypothesis itself uses STA-1 ChIP-seq as the "nematode positive control" for DNA binding — implicitly conceding that equivalent direct DNA-binding data for STA-2 do not exist.

F004 — STA-2 lacks the canonical STAT activating tyrosine (STAT1 Y701) and shows DBD backbone deviation equal to the binding control STA-1

Global BLOSUM62 alignment of STAT1 to each nematode STAT shows that the canonical C-terminal **activating tyrosine STAT1-Y701** — phosphorylated to drive STAT dimerization and high-affinity, sequence-specific DNA binding — aligns to a conserved tyrosine in STA-1 (Tyr588) but has **NO aligned residue in STA-2** (it maps to a gap at the STA-2 C-terminus; STA-2 is truncated at 567 aa, just past its SH2 domain at 449–526). Loss of the phospho-tyrosine switch removes the canonical mechanism by which STATs achieve the stable, reciprocal SH2–phosphotyrosine dimer that clamps onto DNA. This is mechanistically consistent with the "unorthodox" regulation described below.

Alignment-anchored C α superposition of the STAT1 DBD (Pfam region 322–458, PDB 1BF5 chain A) onto the AlphaFold models gives **STA-2 Kabsch RMSD = 8.22 Å over 126 residue pairs** versus **STA-1 = 8.00 Å over 127 pairs** — essentially identical backbone deviation. (These are crude rigid-body RMSDs without outlier trimming and should be read comparatively: STA-2's

fold is no more deviant than the binding-competent STA-1's.) The structural scaffold similarity therefore cannot discriminate binders from non-binders here; the discriminating information lies in the sequence/residue-level and functional data.

F005 — STA-2 is an unorthodox STAT that acts with the SLC6 transporter SNF-12; its DNA engagement may be partner-assisted

Dierking et al. 2011 ([PMID: 21575913](#)) identify the SLC6/solute-carrier transporter **SNF-12** as essential for epidermal AMP induction and show *"the STAT transcription factor-like protein STA-2 as a direct physical interactor of SNF-12,"* with the two proteins *"function[ing] together to regulate AMP gene expression in the epidermis."* The authors explicitly state their findings *"reveal an unorthodox mode of regulation for a STAT factor."* No DNA-binding assay, consensus motif, or ChIP for STA-2 is reported.

Combined with the structural findings — no activating tyrosine, a degenerate DBD that fails PF02864, and loss of all three STAT1 base-contacting residues — this supports a model in which STA-2's transcriptional output is achieved **non-canonically and possibly via protein partners** rather than, or in addition to, autonomous sequence-specific DNA binding.

Mechanistic Model / Interpretation

The evidence converges on STA-2 as a **degenerate / unorthodox STAT** that has retained the STAT architecture as a scaffold while losing the canonical features that make a STAT an autonomous sequence-specific DNA-binding transcription factor.

Canonical STAT (human STAT1)	C. elegans STA-2 (Q20977)
=====	=====
N-domain + coiled-coil (PF01017, PF02865)	[ABSENT] N-terminally truncated (567 aa vs ~706-770 aa)
DNA-binding domain (PF02864) ----DNA base contacts E421/S459/N460 reads GAS/ISRE motif	STAT-DBD FOLD present (pLDDT~88, IPR008967/IPR012345) BUT:
	- FAILS Pfam PF02864
Linker + SH2 domain	- 0/3 base contacts conserved
	- most divergent DBD (20% id)
Activating Tyr (Y701) --(P)--> dimer -> high-affinity DNA binding	
	SH2 domain present (cd09919)
	Activating Tyr ABSENT (truncated)
	Partners with SNF-12 (SLC6)
	tethered at hemidesmosomes;
	released on damage -> nucleus
	-> AMP gene activation
	("unorthodox" STAT regulation)

The functional output (activation of epidermal AMP genes such as *nlp-29*) is genetically real and important for innate immunity, but the *immediate molecular mechanism* by which STA-2 engages chromatin is not established. Three mechanistic possibilities remain open and are not mutually exclusive:

1. **Partner-assisted / tethered DNA engagement** — STA-2 may be recruited to target promoters via SNF-12 or other partners rather than by autonomous sequence reading (most consistent with F001–F005).
2. **Low-specificity or diverged-motif binding** — the retained fold could still contact DNA, but with a motif unlike the mammalian GAS/ISRE and not captured by human-residue conservation metrics (STA-1 also conserves 0/3 human base contacts yet binds DNA).
3. **Non-DNA-binding co-activator role** — STA-2 could act primarily through protein–protein interactions in a transcriptional complex.

For GO curation, the key point is that **GO:0000978 asserts a specific molecular activity (RNA Pol II cis-regulatory region sequence-specific DNA binding) that is currently unproven for STA-2**, while the directly supported activities are (i) a nuclear transcriptional-regulator role in AMP gene expression (BP/MF, genetically inferred) and (ii) a direct physical interaction with SNF-12 (MF: protein binding, experimentally demonstrated).

Evidence Base

Citation (PMID)	Evidence type	Supports / refutes / qualifies	Claim tested	Key finding	Context	Confidence & limitations
InterPro/ Pfam scan (F001)	Structural/ evolutionary; computational	Refutes (functional DBD)	Does STA-2 carry a bona fide STAT DBD signature?	STA-2 fails PF02864 / STAT-DBD CDD that STA-1, STAT1, Stat92E all pass; only fold-level superfamily hits	Sequence analysis, 4 proteins	High for HMM re HMM fail indicates divergen not proo non-bino
1BF5 contact mapping + alignments (F002)	Structural/ evolutionary; computational	Refutes/ qualifies	Are DNA-contacting residues conserved in STA-2?	0/3 base contacts conserved (E421→S, S459→G, N460→P); most divergent DBD (20%); yet AlphaFold fold intact (pLDDT 88)	Human 1BF5 vs AF models	High for conserva gradient human-residue metric imperfec (STA-1 al 0/3)
AlphaFold AF-Q20977-F1 (F002, F004)	Structural; computational	Supports (fold only)	Is the STAT DBD fold retained?	DBD fold predicted at pLDDT 88; DBD backbone RMSD vs 1BF5 = 8.2 Å ≈ STA-1 8.0 Å	In silico model	High-confidence model; RMSD cr compar only
25692704	Mutant phenotype; localization	Qualifies (TF role, no binding assay)	Is STA-2 a transcriptional driver of AMPs?	STA-2 release from hemidesmosomes → AMP transcription / innate immune response	C. elegans epidermis	High for role; no direct DB binding assay
34166401	Mutant phenotype; genetic	Qualifies (TF role, no binding assay)	Is STA-2 a key AMP transcriptional activator?	Fungal effectors block/enhance STA-2 activation of AMP genes;	C. elegans epidermis / fungal infection	High for role; acti genetica inferred

Citation (PMID)	Evidence type	Supports / refutes / qualifies	Claim tested	Key finding	Context	Confidence & limitations
				one raises nuclear STA-2		
21575913	Interaction; genetic	Competing/ qualifies	How does STA-2 regulate transcription mechanistically?	Direct STA-2–SNF-12 (SLC6) interaction; "unorthodox mode of regulation for a STAT factor"	C. elegans epidermis	High for interaction; proposes partner-assisted, non-canonical mechanism

GO Curation Implications

Lead for curator verification — GO:0000978 (RNA Pol II cis-regulatory region sequence-specific DNA binding) on STA-2 should be generalized or removed pending a direct binding assay.

- The term asserts a **specific molecular function (MF)** that is **not experimentally demonstrated** for STA-2 and is contradicted by multiple convergent computational/ evolutionary signals (fails PF02864; loses all STAT1 base contacts; most divergent DBD; no activating tyrosine; no EMSA/ChIP/motif). This is the signature of a **homology-propagated over-annotation**.
- **Recommended action:** Do not retain GO:0000978 with an experimental evidence code. If any annotation to DNA binding is kept, generalize to the less-specific parent **GO:0003677 (DNA binding)** or **GO:0003700 (DNA-binding transcription factor activity)** only with a non-experimental/inferred evidence code (e.g., ISS/IBA) and an explicit caveat, OR remove it as unsupported. A sequence-specificity claim (GO:0000978) is too strong.
- **Better-supported terms to foreground instead:**
- **MF — protein binding (GO:0005515)** is directly supported by the STA-2–SNF-12 interaction (PMID: 21575913). *(Per guidance, this is not a satisfying final MF on its own, but it is the one directly demonstrated molecular activity.)*

- **BP — regulation of antimicrobial peptide / innate immune response gene expression** (e.g., defense response to fungus; regulation of transcription in the innate immune pathway) is supported genetically (PMID: 25692704, PMID: 34166401).
 - **CC — nucleus** (regulated nuclear accumulation on activation) and the hemidesmosome-tethered cytoplasmic pool are both experimentally described.
 - **Net:** Treat STA-2's "sequence-specific DNA binding" as **non-core / unproven**. The core, directly supported story is a damage-released, SNF-12-partnered, nuclear transcriptional regulator of epidermal AMP genes. The DBD is a retained *scaffold*, not a demonstrated *sequence-reading* module.
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Mechanistic Scope

The immediate molecular function under test is **autonomous, sequence-specific binding of STA-2 to RNA Pol II cis-regulatory DNA (GO:0000978)**. The investigation deliberately separates this from downstream and inferred layers:

- **Direct gene-product activity (tested here):** sequence-specific DNA binding — *not demonstrated*; demonstrated direct activity is protein–protein binding to SNF-12.
- **Downstream / pathway consequence:** activation of AMP genes (*nlp-29* and others) — genetically established but one or more steps removed from "does STA-2 touch the DNA sequence-specifically."
- **Developmental/physiological outcome:** epidermal innate immune response to fungal infection and physical damage — a phenotype, not a molecular activity.
- **Loss-of-function inference:** the transcription-factor assignment rests on genetics and localization, i.e., inferred from perturbation and nuclear accumulation, not from a biochemical DNA-binding measurement.

GO:0000978 is a claim at the *first* (direct molecular) level; the strong evidence sits at the *downstream* levels. That mismatch is the crux of the curation decision.

Conflicts and Alternatives

1. **Paralog confusion (STA-1 vs STA-2).** The seed hypothesis uses STA-1 (Q9NAD6, ChIP-validated) as the "nematode positive control." STA-1 passes PF02864; STA-2 does not. Annotations or intuitions transferred from STA-1 (or from mammalian STATs) to STA-2 would constitute paralog over-annotation. The two worm STATs are functionally and structurally distinguishable at exactly the DNA-binding interface.
 2. **Fold $\neq$ function.** AlphaFold fold conservation and near-identical DBD backbone RMSD (8.2 vs 8.0 Å) could be mis-read as evidence of DNA binding. The RMSD metric cannot discriminate STA-2 from the binding-competent STA-1, so it is uninformative for the specific claim and should not be cited as support for GO:0000978.
 3. **Human-residue metric caveat (possible under-call).** STA-1 conserves 0/3 human STAT1 base contacts yet binds DNA in ChIP-seq. This means nematode STATs read DNA through a diverged interface, and "0/3 conserved" alone does not prove STA-2 cannot bind. STA-2's case against binding rests on the *combination* of signals (PF02864 failure + most-divergent DBD + no activating Tyr + no assay), not on residue identity alone.
 4. **Non-canonical mechanism (competing model).** The explicit "unorthodox mode of regulation" and direct SNF-12 interaction ([PMID: 21575913](#)) offer a genuine alternative: partner-assisted recruitment rather than autonomous sequence reading. This is the strongest competing interpretation and is compatible with all structural findings.
 5. **Database carry-over.** GO:0000978 on STA-2 may originate from family-level/electronic annotation of "STAT transcription factor," which would propagate the mammalian sequence-specific DNA-binding activity without worm-specific experimental backing.
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Limitations and Knowledge Gaps

- **No direct DNA-binding assay for STA-2.** *Checked:* primary literature (3 key papers) and the seed context. *Why it matters:* this is the single datum that would decisively confirm or refute GO:0000978. *Resolution:* EMSA with candidate AMP-promoter probes, or ChIP-seq of tagged STA-2 in epidermis.
- **No defined STA-2 binding motif.** *Checked:* literature; none reported. *Why it matters:* "sequence-specific" requires a demonstrable sequence preference. *Resolution:* SELEX/PBM, or motif discovery from STA-2 ChIP-seq peaks.

- **Human-centric contact mapping.** *Checked:* 1BF5 contacts mapped by pairwise alignment. *Why it matters:* nematode STATs use a diverged DNA interface (STA-1 control also 0/3), so residue identity at human positions under-samples true binding determinants. *Resolution:* a nematode STAT–DNA co-structure or STA-1 ChIP motif to define the worm contact code.
 - **Crude RMSD.** *Checked:* rigid-body Kabsch without outlier trimming. *Why it matters:* absolute RMSD (8 Å) is inflated and non-discriminative. *Resolution:* refined structural superposition / local DBD alignment; but note this would not resolve function.
 - **Partner dependency not mapped to chromatin.** *Checked:* SNF-12 interaction demonstrated, but whether SNF-12 (or another factor) provides DNA-contact specificity is unknown. *Resolution:* ChIP of STA-2 with and without SNF-12, and reciprocal.
 - **Activating-tyrosine inference.** *Checked:* alignment shows no aligned Y701 equivalent (STA-2 truncated). *Why it matters:* STA-2 may use a different activation switch; absence of the canonical Tyr does not preclude all DNA binding, only the canonical phospho-dimer route.
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Proposed Follow-up Experiments / Discriminating Tests

The following would most efficiently separate "STA-2 binds DNA sequence-specifically" from "STA-2 is a partner-recruited / non-DNA-binding regulator":

1. **STA-2 ChIP-seq in *C. elegans* epidermis** (ideally comparing basal vs. damage/infection-activated states). Direct occupancy at AMP promoters + a recovered motif would support GO:0000978; absence or exclusively partner-dependent, motif-less occupancy would refute autonomous sequence-specific binding. This mirrors the STA-1 positive-control design in the seed.
2. **In vitro DNA binding (EMSA) of recombinant STA-2 DBD** against candidate AMP-promoter elements and a STAT GAS consensus. Positive binding (with specificity by competition) would rescue the molecular claim despite the degenerate sequence; no binding would corroborate the computational prediction.
3. **PBM / SELEX** on the purified STA-2 DBD to establish whether any sequence preference exists and, if so, how it differs from the GAS/ISRE motif.
4. **SNF-12-dependence test:** STA-2 ChIP in *snf-12* loss-of-function vs. wild type. Loss of occupancy would support a partner-assisted (non-autonomous) model.

5. **Structure-guided mutagenesis:** restore/ablate predicted DNA-contacting positions and assay AMP-reporter activation, to test whether the retained fold's residues matter for function.

Curation Leads (require curator verification)

Candidate action on GO:0000978 (sequence-specific cis-regulatory DNA binding): - **Lead:** Downgrade/remove as an experimentally-coded MF on STA-2; treat as homology-driven over-annotation. If retained at all, generalize to GO:0003700 (DNA-binding transcription factor activity) or GO:0003677 (DNA binding) with a non-experimental evidence code **and a caveat** that direct binding is unproven and sequence-specificity is not demonstrated.

Candidate better-supported terms (leads): - **GO:0005515 protein binding** — directly supported by STA-2–SNF-12 interaction. Candidate reference: [PMID: 21575913](#), snippet to verify: *"the STAT transcription factor-like protein STA-2 as a direct physical interactor of SNF-12."* - **BP: regulation of antimicrobial peptide / innate immune gene expression (defense response to fungus).** Candidate references: [PMID: 25692704](#), snippet: *"detachment of STA-2 molecules from hemidesmosomes and transcription of AMPs"*; [PMID: 34166401](#), snippet: *"preventing the key STAT-like transcription factor STA-2 from activating defensive antimicrobial peptide gene expression."* - **CC: nucleus** (activation-dependent nuclear accumulation) and cytoplasmic/hemidesmosome-tethered pool.

Suggested curator questions: - Is the existing GO:0000978 annotation on STA-2 experimental or electronic/ISS? If electronic family-level, it should not stand as evidence of worm-specific sequence-specific binding. - Was the annotation transferred from STA-1 or mammalian STATs (paralog carry-over)?

Suggested experiments: STA-2 ChIP-seq ($\pm$ SNF-12) and recombinant STA-2 DBD EMSA/PBM, as detailed above.

Evidence snippets to verify (exact, from stored abstracts): - [PMID: 21575913](#): *"reveal an unorthodox mode of regulation for a STAT factor."* - [PMID: 21575913](#): *"the STAT transcription factor-like protein STA-2 as a direct physical interactor of SNF-12."* - [PMID: 25692704](#): *"detachment of STA-2 molecules from hemidesmosomes and transcription of AMPs."* - [PMID: 34166401](#): *"preventing the key STAT-like transcription factor STA-2 from activating defensive antimicrobial peptide gene expression."*

Bottom Line

STA-2 keeps the STAT DNA-binding-domain **fold** (AlphaFold pLDDT $\approx$ 88; DBD backbone RMSD vs. DNA-bound STAT1 1BF5 = 8.2 Å, matching the ChIP-validated binder STA-1 at 8.0 Å), so the hypothesis's first clause holds. But **functional, sequence-specific DNA binding (GO:0000978) is not supported**: STA-2 is the most divergent STAT tested (20% DBD identity), uniquely fails Pfam STAT-DBD PF02864 that STA-1 passes, conserves none of STAT1's three base-contacting residues, lacks the canonical activating tyrosine, and has no direct DNA-binding assay in the literature. The directly demonstrated biology is a damage-released, SNF-12-partnered, nuclear transcriptional regulator of epidermal antimicrobial-peptide genes operating via an explicitly "unorthodox" STAT mechanism. **Curation lead: treat GO:0000978 on STA-2 as homology-driven over-annotation — generalize or remove pending a direct binding assay — while retaining the demonstrated SNF-12 interaction and the AMP/innate-immunity transcriptional-regulation roles.**

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